

Choose Initial Constants

From periodic time series $X_t, t \in \{1, \dots, n\}$ with period, d ,

$$K_n = \lfloor n^{1/3} \rfloor, \lfloor n^{1/2} \rfloor, L_n = \lfloor 4n^{1/4} \rfloor$$

Optimal Block Lengths for Overall Mean

for $K_n \leftarrow K_{n,\min} : K_{n,\max}$ **do**

$$G(K_n) = \sum_{s=1}^d \sum_{k=1}^{K_n} \left\lfloor \frac{k}{d} \right\rfloor \text{Cov}(X_s, X_{s+k}) + \sum_{s=1}^d \sum_{k=1}^{K_n} \left\lfloor \frac{k}{d} \right\rfloor \text{Cov}(X_{s-k}, X_s)$$

end

$$\hat{G}_n = \frac{1}{|K_n|} \sum_{K_n} G(K_n)$$

for $k \leftarrow 1:d$ **do**

$$\hat{f}_k(0) = \frac{1}{2\pi d} \sum_{t=1}^d \sum_{\tau=-L_n}^{L_n} \text{Cov}(X_t, X_{t+\tau}) \exp\left(\frac{-2\pi i k t}{d}\right)$$

end

$$\hat{D}_n = \frac{4}{3} (2\pi d)^2 \sum_{k=0}^{d-1} |\hat{f}_k(0)|^2$$

$$b_{GSBB}^{opt} = b_{CGSBB}^{opt} = \sqrt[3]{\frac{2\hat{G}_n^2}{\hat{D}_n}} \sqrt[3]{n}$$

$$b_{EMBB}^{opt} = b_{CEMBB}^{opt} = \sqrt[3]{\frac{2\hat{G}_n^2}{\hat{D}_n}} \sqrt[3]{n}$$

Optimal Block Lengths for Seasonal Mean

for $K_n \leftarrow K_{n,\min} : K_{n,\max}$ **do**

$$G^s(K_n) = \sum_{k=1}^{K_n} d|k| \text{Cov}(X_s, X_{s+kd}) + \sum_{k=1}^{K_n} d|k| \text{Cov}(X_{s-kd}, X_s)$$

end

$$\hat{G}_n^s = \frac{1}{|K_n|} \sum_{K_n} G^s(K_n)$$

$$\hat{f}_0^s(0) = \frac{1}{2\pi} \sum_{k=1}^{K_n} \text{Cov}(X_s, X_{s+kd}) + \frac{1}{2\pi} \sum_{k=1}^{K_n} \text{Cov}(X_{s-kd}, X_s)$$

$$\hat{D}_n^s(d) = \frac{4}{3} [2\pi d \hat{f}_0^s(0)]^2$$

$$b_{GSBB}^{opt} = b_{CGSBB}^{opt} = \sqrt[3]{\frac{2(\hat{G}_n^s)^2}{\hat{D}_n^s}} \sqrt[3]{n}$$

$$b_{EMBB}^{opt} = b_{CEMBB}^{opt} = \sqrt[3]{\frac{2(\hat{G}_n^s)^2}{\hat{D}_n^s}} \sqrt[3]{n}$$

GSBB: Generalized seasonal block bootstrap
CGSBB: Circular Generalized seasonal block bootstrap
EMBB: Extended moving block bootstrap
CEMBB: Circular extended moving block bootstrap

For all seasonal means:

$$b_{all, GSBB}^{opt} = b_{all, CGSBB}^{opt} = \sqrt[3]{\frac{2 \sum_{s=1}^d (\hat{G}_n^s)^2}{\sum_{s=1}^d \hat{D}_n^s}} \sqrt[3]{n}$$

$$b_{all, EMBB}^{opt} = b_{all, CEMBB}^{opt} = \sqrt[3]{\frac{2 \sum_{s=1}^d (\hat{G}_n^s)^2}{\sum_{s=1}^d \hat{D}_n^s}} \sqrt[3]{n}$$